Niraiva Cancer Patient Workflow

Initial
Cancer workflow begins
Patient Profile
Age gender demographics and observation metadata
Lung Cancer Risk Inputs
Use recorded gender smoking status and quit-smoking age when applicable
Lifestyle and Exposure Assessment
Smoking alcohol diet obesity physical activity and UV exposure
Alcohol Consumption
Alcohol status; type; units per week; frequency; duration
Reported value: ~4-5% cancer attribution; cited male fractions ~25-50% for several cancers
Diet and Nutrition
Dietary pattern; fruit; vegetables; fibre; whole grains; processed foods
Obesity and Body Weight
BMI; body weight; waist measurement; overweight or obesity status; weight history
Reported value: Esophageal: 29% men, 37% women; endometrial: ~36%; gallbladder: 11% men, 42% women
Red and Processed Meat
Red meat and processed meat consumption; servings and quantity per week
Reported value: Recommended maximum: 18 oz / ~510 g / 3-4 servings per week
Environmental Exposures
Air and water quality; inorganic arsenic; occupational glyphosate, malathion and diazinon
Combined Lifestyle Impact
About 18% of cancer cases and 16% of cancer deaths in one reviewed estimate
Food Evidence and Cautions
High-temperature cooking; acrylamide uncertainty; glycemic load; very hot drinks; coffee; GM, irradiated food, microwave, supplements and detox claims
Lifestyle Biological Pathways
DNA damage; acetaldehyde; oxidative stress; inflammation; hormonal/metabolic/immune and microbiome pathways; tobacco-alcohol and alcohol-hepatitis interactions
Smoking and Tobacco
Smoking status; tobacco exposure; duration; frequency; cigarettes per day
Reported value: >80% cited lung-cancer attribution; ~10-20x lung-cancer risk
Ultraviolet Exposure
Sun exposure; UV exposure; occupation; tanning exposure; duration and protection
Physical Activity
Activity type; frequency; intensity; duration; minutes per week and sedentary behaviour
Reported value: Up to ~50% lower colorectal risk; 48% lower among men in one cited study
Laryngeal Smoking Value
Smoking-attributable fraction exceeds 50% in the cited analysis
UV Reported Value
About 90% of skin cancers attributed to UV exposure in the cited source
Additional Activity Finding
Initially inactive men increasing activity: about 13% lower reported risk
Gender for Lung Risk Model
Select the sex-specific probability branch recorded in the source model
Male
Female
Male Smoking Status
Current smoker former smoker or never smoker
Female Smoking Status
Current smoker former smoker or never smoker
Former smoker
Current smoker
Never smoker
Quit-Smoking Age - Male Branch
Record cessation age for former smokers or mark it unavailable
Former smoker
Never smoker
Current smoker
Quit-Smoking Age - Female Branch
Record cessation age for former smokers or mark it unavailable
Age recorded / unavailable
Male Lung Risk Calculation
Apply male smoking status and quit-smoking age when applicable
Age recorded / unavailable
Female Lung Risk Calculation
Apply female smoking status and quit-smoking age when applicable
Probability result
Male Probability Values
Smoker 0.244 (24.4%); quit smoker 0.055 (5.5%); nonsmoker value unavailable in source
Probability result
Female Probability Values
Smoker 0.185 (18.5%); quit smoker 0.026 (2.6%); nonsmoker 0.004 (0.4%)
Lifestyle Risk Summary
Combine assessed lifestyle and exposure factors to inform risk-aware screening and clinical review
Concerning Symptoms or Clinical Changes?
Select screening or diagnostic path
No - Routine screening
Yes - Symptoms present
Cancer-Specific Screening
Screening where applicable
Diagnostic Clinical Encounter
Clinical encounter for symptoms or findings
Suspicious Clinical Finding?
Determine whether further investigation is required
No - Routine monitoring
Yes - Investigate
Diagnostic Work-up
Laboratory and imaging assessment
Laboratory tests
Imaging
Laboratory Assessment
CBC CMP CRP and ESR measurements
Imaging Assessment
Cancer-specific imaging where applicable
CBC Assessment
Hemoglobin; RBC; WBC; platelets; RBC-related indices; WBC differential
CMP Assessment
ALP; AST; ALT; calcium; albumin; bilirubin; other metabolic measurements
Inflammation Assessment
CRP and ESR; previous value; current value; date and trend
Suspicious Lesion or Abnormality?
Review clinical imaging and laboratory findings
Imaging Method
Mammography; CT; MRI; ultrasound or other applicable imaging
Lesion Characteristics
Presence; location; size; shape; margins; count
Spread Assessment
Regional lymph-node findings and distant findings
No - Reassess
Yes - Biopsy
Biopsy or Tissue Sampling
Tissue sampling where clinically applicable
Biopsy Details
Biopsy method; date; anatomical site; specimen
Pathology Assessment
Histology tumor classification and pathology observations
Histopathology
Histology; tumour classification; microscopic observations
Routine Monitoring
Monitoring when no suspicious finding is identified
Monitor and Reassess
Repeat assessment when appropriate
Malignancy Confirmed?
Pathology confirmation decision
No - Non-malignant
Yes - Malignant
Cancer Type Classification
Cancer-specific classification
Non-Malignant Follow-up
Non-cancer follow-up or general care when malignancy is not confirmed
Breast Cancer Pathway?
Determine whether breast-specific receptor data apply
Yes - Breast pathway
No - Other cancer
HER2 ER and PR Assessment
Breast-specific receptor assessment where applicable
Estrogen Receptor
ER staining result; positive percentage; final ER status
HER2 Assessment
IHC result; FISH result; ERBB2 signal; ratio; final HER2 status
Progesterone Receptor
PR staining result; positive percentage; final PR status
TNM Assessment
Assessment of primary tumor regional nodes and distant metastasis
N
T
M
Alternative Staging Systems
Lugano for selected lymphomas; FIGO for female reproductive cancers; conventional TNM may not apply to leukemia or many brain/CNS tumours
Cancer-Specific Staging Caution
Stage 0 and TNM groupings are cancer-specific; map M1 to Stage IV only where the applicable cancer-specific system defines it
N - Regional Lymph Node Assessment
Regional lymph-node involvement
T - Primary Tumor Assessment
Tumor size depth and local extension
Grade Profile and Prognosis
Stage is anatomical extent; grade is differentiation; profile includes histology, biomarkers and molecular/genetic features; stage is not individual prognosis
Clinical TNM cTNM
Clinical stage established before definitive treatment using clinical examination and diagnostic investigations
Cancer Staging Overview
Cancer staging describes the anatomical extent of disease using clinical and pathological information
Recorded TNM Clinical Fields
Primary tumour size/local extension; involved-node count; micro/macrometastasis; metastasis location
Post-Treatment and Restaging
yTNM after specified preoperative treatment; rTNM for recurrence; retain original diagnosis stage
Stage Subdivisions and SEER Extent
IIA/IIB; IIIA/IIIB/IIIC; SEER In situ, Localized, Regional, Distant and Unknown
Staging Investigations
Physical examination; CT; MRI; ultrasound; PET/PET-CT; endoscopy; biopsy and selected laboratory tests
M - Distant Metastasis Assessment
Presence or absence of distant spread
Pathological TNM pTNM
Pathological stage established using preoperative evidence plus surgical and histopathological findings
N Regional Lymph Nodes
N describes involvement of regional lymph nodes
T Primary Tumour
T describes the size or direct extent of the primary tumour
TNM Category Integration
Combine the assessed T N and M categories before cancer-specific stage grouping
N1 through N3
Increasing regional lymph-node involvement; exact definitions depend on cancer type
N0
No regional lymph-node metastasis
T1 through T4
Increasing tumour size or increasing local extent; exact definitions depend on cancer type
Integrated T / N / M
Distant Metastasis Identified?
M1 or M0 branch
M Distant Metastasis
M describes the presence or absence of distant metastasis
General care / close pathway
M0 - No metastasis
M1 - Metastasis
M0
No distant metastasis identified
M1
Distant metastasis is present
In Situ Disease?
Tis where Stage 0 applies
Tis - In situ
Not in situ
Tis Carcinoma In Situ
Carcinoma in situ without invasion of surrounding tissues
Stage IV - Metastatic Disease
Distant or metastatic disease
Cancer-Specific Stage Group
Combine applicable T N and M categories
Stage III
Stage I
Stage II
Stage III - Regional Disease
Regional or locally advanced disease
Stage I - Localized
Localized or early disease
Stage II - More Locally Advanced
Greater local disease extent
Stage 0 - In Situ (Where Applicable)
Carcinoma in situ without invasion; applicability and definition vary by cancer type
Return to surveillance
Multidisciplinary Treatment Planning
Stage, profile/grade, patient factors and preferences guide surgery, radiation, chemotherapy, immunotherapy, targeted/hormone/systemic therapy and supportive or palliative care
Follow-up and Surveillance
Symptoms laboratory tests imaging lifestyle and well-being
Progression or New Finding?
A progression concern or new finding returns to clinical assessment and diagnostic confirmation before any restaging
No - End workflow
Yes - assess and confirm
Module End
End of current workflow cycle
Clinical Assessment
Assessment of systemic surface and functional changes
Breast Clinical Signs
Lump or thickening; location; size; skin or shape change; nipple change; discharge; pain; duration; progression
Legend - Color and Line Meaning
Clinical process
Start / end / major state
Decision / lifestyle risk
Laboratory / imaging
Biopsy / pathology
Biomarkers / receptors
Monitoring / protective
TNM staging
Cancer stage group
Advanced / metastatic
Reported value / note
Lines
Solid: workflow; Dashed: detail